Pre-Stall Load Redistribution on Swept Tailplane Models with Full-Span Controls

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Local loading changes on a swept lifting surface with a deflected plain control can alter lift growth and pitching moment well before maximum lift. MONNALISA wind-tunnel (WT) measurements and Reynolds-averaged Navier–Stokes (RANS) solutions are used to examine how fixed-surface and full-span control loads combine in this response. For the positive-sweep, zero-dihedral model at a 20° deflection, the computed fixed surface continues gaining lift while both surface contributions fall below their preceding incidence trends. Across the tested baseline windows and zero-incidence solutions, the fixed surface accounts for about 63% of the total lift departure; the control contributes more strongly to the moment departure about the fitted aerodynamic center. Pressure sections and regional integrations connect these different allocations to stronger leading-edge suction, reduced hinge/aft loading, and larger outboard moment weights. Matched dihedral comparisons distinguish absolute load projection from changes with incidence. The detailed interpretation concerns the computed states, for which correspondence with measured integrated coefficients is partial. The combined force and moment analysis identifies loading changes that continuing total lift growth alone would conceal.

Nomenclature

C_D, C_L	=	drag and lift coefficients
$C_{F,k}$	=	body-force coefficient, $k = x, y, z$
$C_M^A, C_M^O, C_M^{A_I}$	=	moments about fitted center, origin, or common configuration-I axis
$C_p,\ \mathbf{C}_f, C_{f,c}$	=	pressure, vector skin-friction, and chordwise skin-friction coefficients
$D_{Q,j}$	=	departure from the preceding coefficient trend
F_k	=	body-force component, N

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$\mathcal{I}_M$	=	dihedral contrast of moment increments
M_∞	=	freestream Mach number
Q	=	coefficient label, L or M
$Re_b, Re_{\bar{c}}$	=	semispan and mean-chord Reynolds numbers
S	=	undeformed semiplanform reference area, m^2
T_∞	=	freestream temperature, K
U_∞	=	freestream speed, m/s
b	=	undeformed semispan, m
$c,\ \bar{c}$	=	local and mean aerodynamic chords, m
f_{rev}	=	upper-aft area fraction with reverse chordwise wall shear
i, j, k	=	sample, surface-contribution, and force-component indices
$m^{\mathrm{pre}}_{Q,j}$	=	preceding slope, per degree
p, p_∞	=	surface and freestream pressures, Pa
q_∞	=	freestream dynamic pressure, Pa
s	=	undeformed span coordinate, m
t	=	section thickness, m
$\mathbf{t}_c$	=	downstream chordwise unit tangent
u_i, w_i	=	supplied sample uncertainty magnitude and contrast weight
$u_{\mathrm{ind}}, u_{\mathrm{L1}}$	=	independence and absolute-weight contrast sensitivities
x, y, z	=	downstream, spanwise, and upward body coordinates, m
y^+	=	nondimensional wall distance
x_A, x_O	=	fitted-center and origin abscissae, m
$\alpha, \alpha_a, \alpha_b$	=	incidence and interval endpoints, deg
Γ, δ	=	dihedral and control deflection, deg
$\Delta C_{Q,j}$	=	ordinary coefficient increment
γ_{BC}	=	algebraic transition-model intermittency factor
ϵ	=	reversed-shear threshold

$\eta = s/b,\ \xi$ = span fraction and straightened local chord fraction

λ, Λ = quarter-chord sweep (deg) and aspect ratio

ρ_∞ = freestream density, kg/m^3

$\tau, \boldsymbol{\tau}_w$ = taper ratio and wall-shear vector (Pa)

I. Introduction

A deflected trailing-edge control changes both the amount of lift generated by a lifting surface and the location at which that load acts. Its response with incidence is therefore relevant to pitching moment as well as lift, even before the maximum-lift condition is reached. When local loading changes, total lift can continue increasing while its growth slows and the pitching response changes more abruptly. Interpreting this behavior requires relating the integrated coefficients to the loads on the fixed and moving parts of the surface.

Sectional plain-flap experiments establish the starting point for this problem. Sears and Liddell [1] measured lift, drag, pitching moment, and hinge moment for a 30%-chord plain flap on an NACA 0015 airfoil, including sealed and unsealed hinge gaps. Separation reduced flap effectiveness at large deflections, although additional deflection still increased lift within the tested range. Their comparison with a thinner symmetric section also showed that similar lift effectiveness could coexist with different hinge-moment behavior. Reduced flap effectiveness is thus established physics; neither the presence of separation nor a single integrated lift trend fully characterizes the loading response.

On a finite wing, flap-induced loading is also shaped by the spanwise flow and planform. Campbell, Blanks, and Leaver [2] investigated rectangular wings of small aspect ratio with full-span plain flaps of several chord fractions. Their force and pitching-moment measurements, compared with thin-wing theory and empirical thickness and boundary-layer allowances, related flap effectiveness and center of pressure to flap chord and aspect ratio. Those unswept, thinner, transition-tripped models differ from the present swept models, but they establish why the location of the additional load matters alongside its magnitude.

For swept wings, Harper and Maki [3] documented how spanwise changes in local lift response can produce pitching-moment nonlinearity below maximum lift. The first appearance of local stall need not coincide with the first integrated moment change, because the force change and its moment arm vary along the span. Their broad stall review provides context for examining load redistribution, rather than an equivalent test of the present deflected-control configurations.

Nonlinear sectional data provide one route for representing these responses in reduced-order models. Sivells and Neely [4] coupled such lift data to lifting-line theory and also calculated wing pitching moment from sectional moment data. Goitia and Llamas [5] coupled a vortex-lattice formulation to viscous sectional data, with effective-sweep and finite-wing corrections, to predict nonlinear wing and tail lift. Consequently, the ability of a reduced-order model to

reproduce the response depends on how sectional nonlinearities and three-dimensional loading changes are represented.

The immediate basis for the present work is the MONNALISA database publication [6]. Its Sec. IV and Fig. 11 already document the baseline pre-stall lift nonlinearity, expansion of reversed flow on the control, associated pressure changes, and continued growth of total lift. Its DUST formulation uses sectional tables and corrections, while the retained potential-based panel and vortex-lattice predictions are smoother. The present loading analysis builds on these established observations.

The remaining loading question is how changes on the fixed surface and control combine to produce the joint lift and moment response. Continued fixed-surface lift growth, for example, does not establish that this contribution follows its preceding trend. Comparing ordinary increments with departures from that trend identifies which components supply the reduced total growth, while spatial pressure integrations show why the component ranking can differ for moment. This study develops that quantitative interpretation from the existing Reynolds-averaged Navier–Stokes (RANS) solutions, first assessing their integrated coefficients against the measured response and its supplied uncertainty. Changes in control deflection and geometry then test how broadly the reference loading pattern persists within the dataset.

The test articles are isolated tailplane-inspired wings with full-span, 30%-chord plain controls; no fuselage, wing wake, or complete-aircraft trim condition is represented. The 45° dihedral setting supplies a pronounced orientation contrast on the same planform, while a forward-swept planform provides the contrasting sweep direction. These comparisons define the scope of the analysis without establishing behavior for partial-span controls, other flap chords, or intermediate dihedral angles.

II. Experimental, Numerical, and Analysis Framework

A. Models and experimental installation

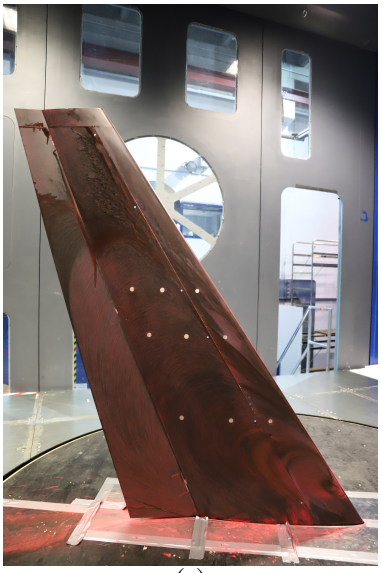
Wind-tunnel (WT) measurements establish the integrated lift and pitching-moment response, while RANS surface loads identify how the fixed surface and control contribute to it. To examine dihedral, the reference aft-swept planform (configuration I) is investigated at $\Gamma = 0°$ and 45°, with its in-plane geometry unchanged. A forward-swept planform (configuration II) provides a contrast in sweep direction and overall planform response. The labels I and II are used in the tables.

Both models use a symmetric section of constant profile along the undeformed semispan $b = 1.6$ m, with thickness ratio $t/c = 9\%$, where c is the local chord. Each model has a full-span plain control hinged at 70% chord. The reference surface S is the undeformed semiplanform area, and $\bar{c}$ denotes its mean aerodynamic chord (MAC). Table 1 gives the geometric parameters and coefficient references. The quarter-chord sweep λ is +30° for configuration I and −15° for configuration II; positive dihedral Γ raises the tip. Aspect ratio is defined for the corresponding full planform as $\Lambda = 2b^2/S$, and τ is the tip-to-root chord ratio. Configuration II also has a different taper and aspect ratio, so the sweep

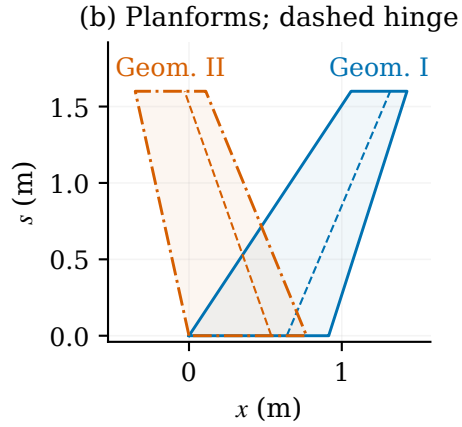
contrast is interpreted through the overall planform response.

Table 1 Model geometry and coefficient reference quantities.

Property	I, $\Gamma = 0°$	I, $\Gamma = 45°$	II, $\Gamma = 0°$
λ (deg)	30	30	−15
Λ	5.0	5.0	5.2
τ	0.4	0.4	0.6
S (m^2)	1.024	1.024	0.984604
$\bar{c}$ (m)	0.679184	0.679184	0.6282
Root chord (m, rounded)	0.914286	0.914286	0.76923
$Re_{\bar{c}}$	2.335×10^6	2.335×10^6	2.159×10^6
Numerical x_A (m)	0.642000	0.643532	−0.003618



(a)



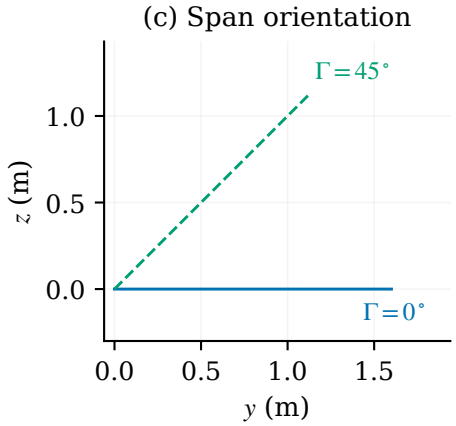


Fig. 1 Wind-tunnel installation and geometry: (a) mounted semispan model; (b) reference and forward-swept planforms; (c) zero- and positive-dihedral arrangements.

The measurements were obtained in the low-turbulence section of the Galleria del Vento del Politecnico di Milano (GVPM), a closed-circuit facility. The test section is 4.00 m wide and 6.00 m long, with an original height of 3.84 m.

For the present tests, installation of the false floor reduced the test-section height to 3.49 m; the undeformed model semispan is 40% of its width. A curved upstream inlet and gap beneath the false floor were designed to divert the incoming tunnel-floor boundary layer below the model installation. Boundary-layer development over the raised floor and around the support remains distinct from a numerical symmetry boundary. The installation photograph in Fig. 1 was taken at $\alpha = 25°$, $\delta = 20°$ and illustrates the mounting, rather than the pre-stall state.

The database setup description [6] gives freestream turbulence below 0.1%, measured using a turbulence sphere and hot-wire anemometry. Outside the wall layers, the inlet total-pressure variation was within ±0.5% of dynamic pressure over most of the section and within ±1% overall; outlet measurements confirmed similar uniformity. The reported longitudinal static-pressure gradient is $\partial C_P/\partial x = -0.004\ \mathrm{m}^{-1}$. Loads were acquired through a RUAG 767 strain-gauge balance below the floor and an HBM QuantumX system; the reduction includes the balance-to-wind transformation and the database corrections. The nominal freestream Mach number and temperature are $M_\infty = 0.147$ and $T_\infty = 300$ K, with MAC Reynolds numbers given in Table 1. The common database operating point is $Re_b = 5.5 \times 10^6$; the MAC-based values are $Re_{\bar{c}} = Re_b \bar{c}/b$.

Recorded incidences are $-25, -24, -22, \ldots, 22, 24, 25°$: the local pre-stall spacing is 2°, with 1° endpoint steps. The comparisons use $\delta = 0, 10, 20, 30°$. Uncertainty bars are the supplied propagated measurement/reduction-chain magnitudes, including balance and transformation contributions. Each fitted-center moment retains its associated propagated uncertainty. A common coverage factor and between-incidence covariance are unavailable; the experimental contrast sensitivities below state their assumptions explicitly.

B. Conventions and moment references

Positive incidence α raises the leading edge relative to the freestream, and positive control deflection δ moves the trailing edge downward relative to the fixed section. Lift is positive toward the upper surface and pitching moment is positive nose-up. The WT and numerical data are expressed in these common physical conventions. In body axes, x is downstream, y is spanwise, and z is upward. The body-force components F_k, with $k = x, y, z$, are normalized by the freestream dynamic pressure $q_\infty = \rho_\infty U_\infty^2/2$ and area S, where ρ_∞ and U_∞ are freestream density and speed. Lift and drag coefficients follow from

$$C_L = C_{F,z}\cos\alpha - C_{F,x}\sin\alpha, \qquad C_D = C_{F,z}\sin\alpha + C_{F,x}\cos\alpha, \qquad C_{F,k} = F_k/(q_\infty S). \tag{1}$$

Pitching moment is normalized by $q_\infty S\bar{c}$. Body coordinates are expressed relative to the root leading edge.

Both experimental and numerical moments are referred to aerodynamic centers determined from the zero-deflection loading characteristics. Linear fits to moment and body-normal force over $|\alpha| \le 10°$ define the positions, which remain fixed when control deflection varies [6]. The experimental fits use the supplied uncertainty weights; the numerical fits

use the retained zero-deflection linear trends. For configuration I, the WT and RANS positions are 0.6430 and 0.6420 m at zero dihedral, and 0.6455 and 0.6435 m at 45°: separations of about 1.0 and 2.0 mm. For configuration II they are −0.0492 and −0.00362 m, separated by 45.6 mm. The comparison coefficient C_M^A therefore uses the same reference construction, with each dataset's fitted position; it is not an imposed quarter-MAC moment.

The original numerical reference is $O = (0.228571, 0, 0)$ m, the root quarter-chord of Geom. I. For a numerical pole transported from abscissa x_0 to x_1,

$$C_M(x_1) = C_M(x_0) + \frac{x_1 - x_0}{\bar{c}} C_{F,z}. \tag{2}$$

The fitted abscissae are in Table 1. Matched dihedral diagnostics use the identical numerical axis $A_I = (0.642000, 0, 0)$ m in both configurations, denoted $C_M^{A_I}$. Geom. II uses its own MAC once for every moment contribution; all coefficients use the fixed areas in Table 1.

C. Numerical method and existing accuracy evidence

The SU2 [7] calculations solve compressible ideal-gas RANS with Sutherland viscosity, the Spalart–Allmaras (SA) turbulence model [8], and the Bas–Cakmakcioglu (BCM) algebraic transition treatment [9]. The finite-volume mean-flow discretization uses median-dual vertex cells and Roe fluxes with second-order monotonic upstream-centered schemes for conservation laws (MUSCL) reconstruction [10], Green–Gauss gradients, and a Venkatakrishnan limiter [11] with coefficient 0.05. SA convection uses first-order scalar upwinding. The BCM intermittency factor γ_{BC} modifies turbulence production without an additional transported transition equation.

The model surfaces have no-slip, adiabatic boundaries, with freestream conditions at the far field and symmetry at the root. The symmetry plane omits the physical support and boundary-layer development over the raised-floor arrangement. Steady solutions use implicit Euler pseudo-time advancement and flexible generalized minimal residual (FGMRES) linear solves, with adaptive Courant–Friedrichs–Lewy (CFL) number. The production stopping criterion monitors lift and drag over 100 entries with a Cauchy tolerance of approximately 3×10^{-6}. Case-specific solution sensitivity is discussed where it affects the interpretation.

The surface grid resolves the leading edge, hinge junction, and trailing edge, where curvature and pressure changes require additional resolution. Wall-normal boundary-layer extrusion uses a nominal first-cell height of about 2×10^{-6} m and growth ratio 1.15, while incidence-aware volume refinement follows the wake. The volume mesh is hex-dominant and octree-based [6]. Representative solutions have median nondimensional wall distance $y^+ = 0.19$–0.26; local maxima are about 1.1–1.3, so the median does not imply that every point has $y^+ < 1$. Production grids contain on the order of 10^7 volume points. A reference grid contains approximately 9.44 million points and 18.92 million volume elements.

The existing grid-refinement assessment uses four levels—coarse, nominal, fine, and very fine—for configuration I at $\delta = 35°$, $\Gamma = 0, 45°$, and $\alpha = 0, 18°$. Table 2 gives the point and element ranges. Figure 2 compares lift, drag, and

pitching moment against the actual grid-point count on a logarithmic scale; open squares identify the nominal level, and lines connect the retained levels. Moments are about the numerical origin O, using the configuration-I area and MAC.

The nominal level was selected to balance spatial resolution with the demands of the extensive MONNALISA campaign across geometries, incidences, and control deflections. Its approximately 9.7–12.7 million points compare with 50.1–65.9 million on the finest level. Nominal-to-finest differences are 1.20–2.14% for lift and 1.13–2.49% for drag. At zero dihedral and $\alpha = 18^\circ$, C_M^O changes from approximately -0.813 to -0.827, an absolute difference of 0.0134; the inclined moment sequence is nonmonotonic. The assessment supports the nominal grid as a practical production compromise while showing finite, condition-dependent integrated-load sensitivity. It provides no discretization-error bound for local component loads. The comparison with measurements in Sec. III separately assesses reproduction of the principal pre-stall aerodynamic trends.

Table 2 Point and volume-element ranges, in millions, for the four-level refinement assessment.

Refinement	Points	Volume elements
Coarse	4.79–5.29	9.51–10.90
Nominal	9.67–12.74	19.06–24.07
Fine	22.33–26.56	39.93–45.70
Very fine	50.11–65.89	87.35–109.98

D. Additive loading and empirical departures

A total lift increase can conceal reduced growth of one or more surface contributions. The analysis therefore compares the actual coefficient change with the change obtained by continuing an empirical preceding trend, then resolves both measures into additive loads. Solver surface groups define the fixed surface (upper/lower fixed panels), the control (upper/lower control panels, hinge junction, and blunt trailing edge), and tip. Pressure and viscous contributions are retained on both physical sides with the same global S and $\bar{c}$. Tip is included in every total, and all component coefficients are displayed at their actual magnitude.

For coefficient label $Q = L$ or M and surface contribution j (fixed, control, tip, or total), the ordinary increment is the change between sampled incidences α_a and α_b. Its departure subtracts the change predicted by continuing the preceding slope $m^{\mathrm{pre}}_{Q,j}$, measured per degree,

$$\Delta C_{Q,j} = C_{Q,j}(\alpha_b) - C_{Q,j}(\alpha_a), \qquad D_{Q,j} = \Delta C_{Q,j} - m^{\mathrm{pre}}_{Q,j}(\alpha_b - \alpha_a). \tag{3}$$

The preceding slope is an unweighted linear least-squares fit to the declared preceding samples. Using identical samples and weights for every component and the total preserves additivity for both increments and departures. The primary preceding-incidence range for $\delta = 20^\circ$ is $\{-6, -4, -2\}^\circ$; the early reference $\delta = 30^\circ$ numerical baseline is

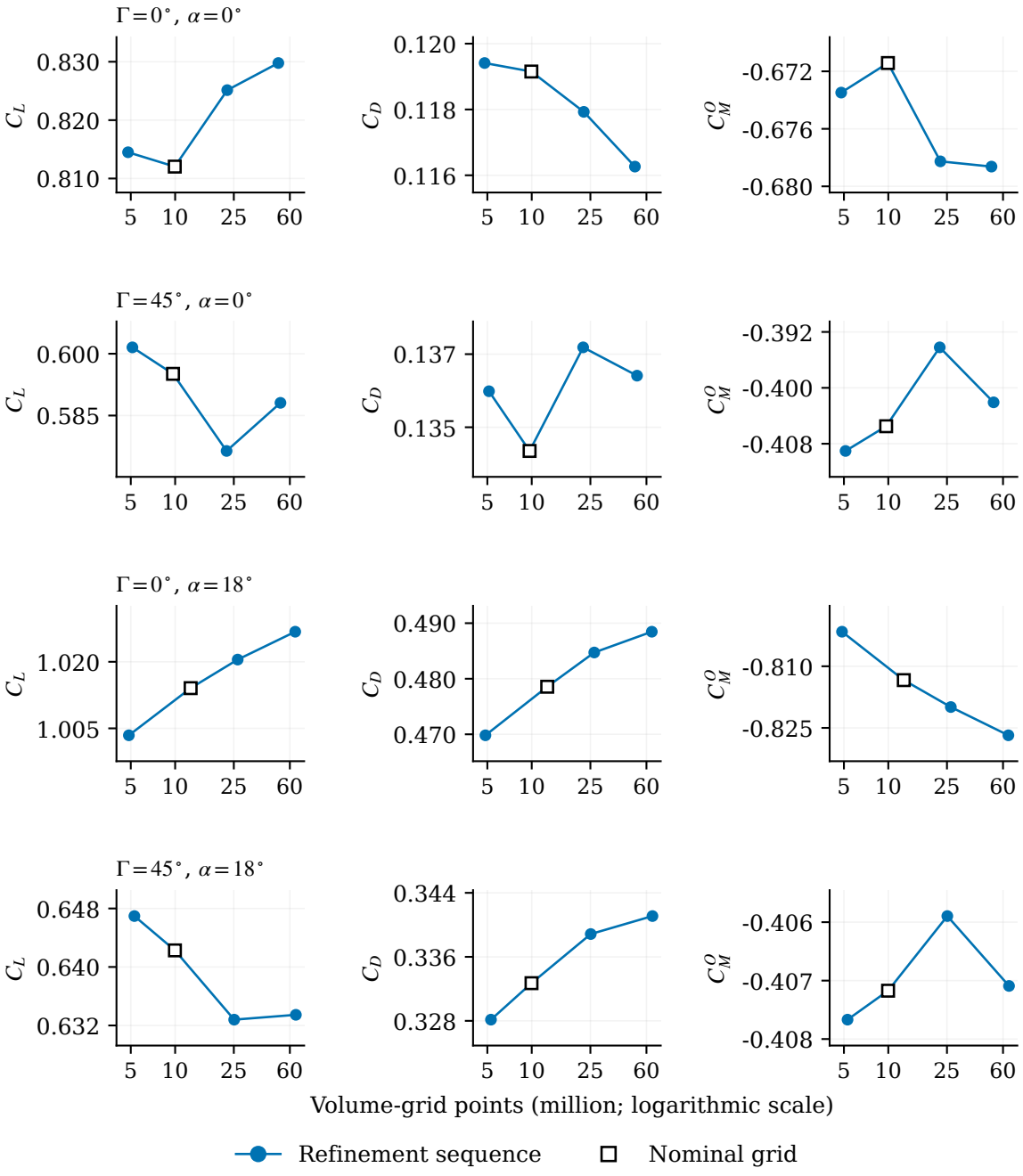


Fig. 2 Grid-refinement trends of lift, drag, and pitching moment for configuration I at $\delta = 35°$, two incidences, and two dihedral settings.

$\{-16, -14, -12\}^\circ$. These are empirical references, with no assumption that the preceding flow is attached. Neighboring preceding-incidence fitting ranges and alternate zero-incidence solutions test this reference dependence (Appendix A). Where the preceding sequence is curved, signed finite differences describe the response without imposing a unique event amplitude.

Measured departures must also be assessed against the supplied uncertainty. For WT contrasts, Eq. 3 is written as $D = \sum_i w_i C_i$, where i indexes the measured samples, w_i is the combined coefficient weight, and u_i is its supplied propagated uncertainty magnitude. The independence-based sensitivity u_{ind} and absolute-weight sensitivity u_{L1} are

$$u_{\text{ind}} = \left[\sum_i (w_i u_i)^2\right]^{1/2}, \qquad u_{\text{L1}} = \sum_i |w_i| u_i. \tag{4}$$

The first is conditional on independence; the second bounds a perturbation when each sample varies within its supplied magnitude. Neither is an established confidence interval or a new experimental uncertainty budget. Covariance-dependent significance is not inferred.

To locate the changes identified by the component budget, pressure and wall-shear forces are integrated over chordwise and spanwise regions. The pressure coefficient is $C_p = (p - p_\infty)/q_\infty$, and the skin-friction vector is $\mathbf{C}_f = \boldsymbol{\tau}_w / q_\infty$, where p_∞ is freestream pressure and $\boldsymbol{\tau}_w$ is wall shear. Linear surface tractions and their moments are integrated over the triangulated surface, with

$$C_M(x_A) = \frac{1}{q_\infty S \bar{c}} \int \left[z\, dF_x - (x - x_A) dF_z \right]. \tag{5}$$

Here x_A is the moment-reference abscissa, and dF_x, dF_z are local force contributions. The axial-force term $z\, dF_x$ and normal-force term $-(x - x_A) dF_z$ are retained separately for the common-axis dihedral comparison. Reconstructed full-surface forces and moments are checked against the solver-integrated coefficients before regional partitioning. Their maximum lift and moment differences are below 10^{-4}; the detailed closure and effect-scale checks remain in the reproducibility records.

The regions distinguish loading upstream of the hinge from loading on the deflected control, and compare inboard, middle, and outboard contributions. The nondimensional span position is $\eta = s/b$, where s is distance along the undeformed span; ξ is the local chord fraction after straightening the control about its hinge. Chordwise regions are forward ($\xi < 0.4$), intermediate ($0.4 \le \xi < 0.7$), and aft ($\xi \ge 0.7$). Spanwise regions divide the semispan into thirds. Contributions include both physical sides, with end-face loads completing the total. Sectional C_p is shown at $\eta = 0.2, 0.5, 0.8$ on both sides of the fixed and control panels, excluding the hinge junction and end faces. Regional contributions are integrated loads, rather than load densities.

Local reverse flow is identified by wall shear directed upstream. We report the fraction f_{rev} of the actual upper

aft-surface area ($\xi \geq 0.7$, excluding end faces) for which $\mathbf{C}_f \cdot \mathbf{t}_c < -\epsilon$. The signed chordwise coefficient is $C_{f,c} = \mathbf{C}_f \cdot \mathbf{t}_c$, where the unit vector $\mathbf{t}_c$ points downstream along the local chord in the surface tangent plane; $\epsilon = 0$ gives the primary definition. Small positive thresholds test near-zero sensitivity in the appendix. This indicator quantifies reverse chordwise wall shear, rather than the complete separated area in three-dimensional flow.

III. Measured Response and Numerical Correspondence

The experimental comparison establishes which lift and moment changes occur together, then assesses how RANS reproduces them. Figure 3 presents the reference zero-dihedral polars for four deflections. Figures 4, 5, and 6 compare WT and RANS for the reference, inclined, and forward-swept configurations, respectively. Filled/solid curves denote WT and open/dashed curves RANS; circles and squares distinguish 20° and 30° deflections, and open diamonds mark alternate zero-incidence solutions. Strips below each plot locate selected WT intervals, whose departures and uncertainty sensitivities are assessed in Table 3. Moments use each dataset's fitted aerodynamic center.

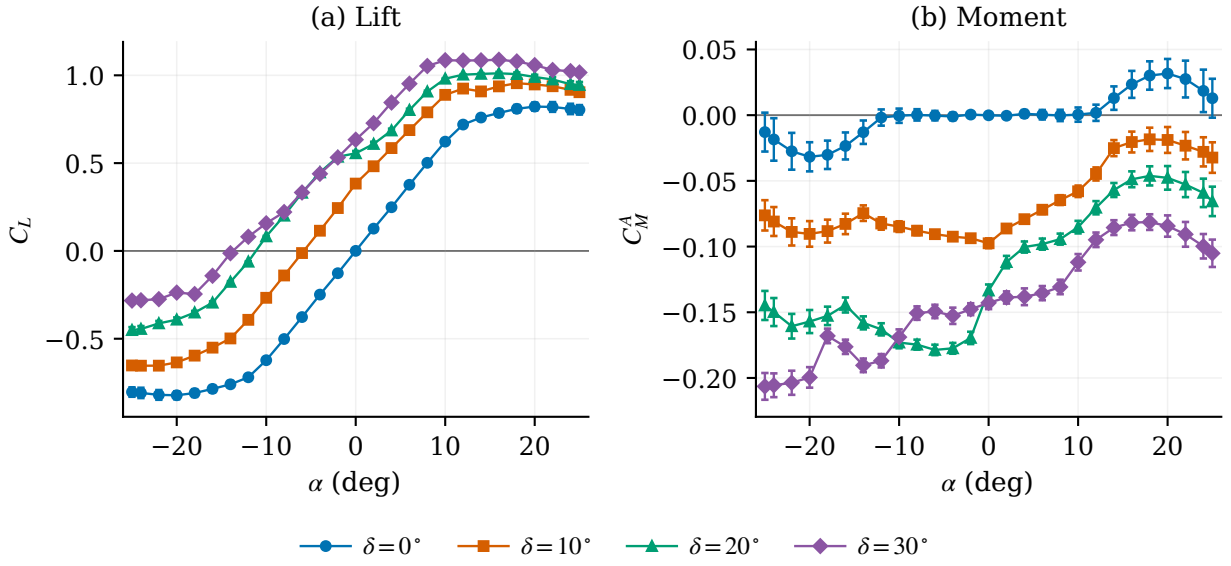


Fig. 3 Experimental lift and pitching moment of the reference planform at zero dihedral for four control-surface deflections.

For the zero-dihedral reference configuration at $\delta = 20°$, the lift increment over $-2 \rightarrow 0°$ is approximately 78% smaller than over $-4 \rightarrow -2°$, although lift continues increasing. The respective increments are about 0.020 and 0.089. Pitching moment increases by +0.036, becoming less negative while retaining its sign. Both departures in Table 3, calculated against $\{-6, -4, -2\}°$, exceed the stated uncertainty sensitivities; the corresponding pointwise lift- and moment-coefficient uncertainty magnitudes are approximately 0.013 and 0.0046–0.0047. These are changes over the available 2° incidence spacing.

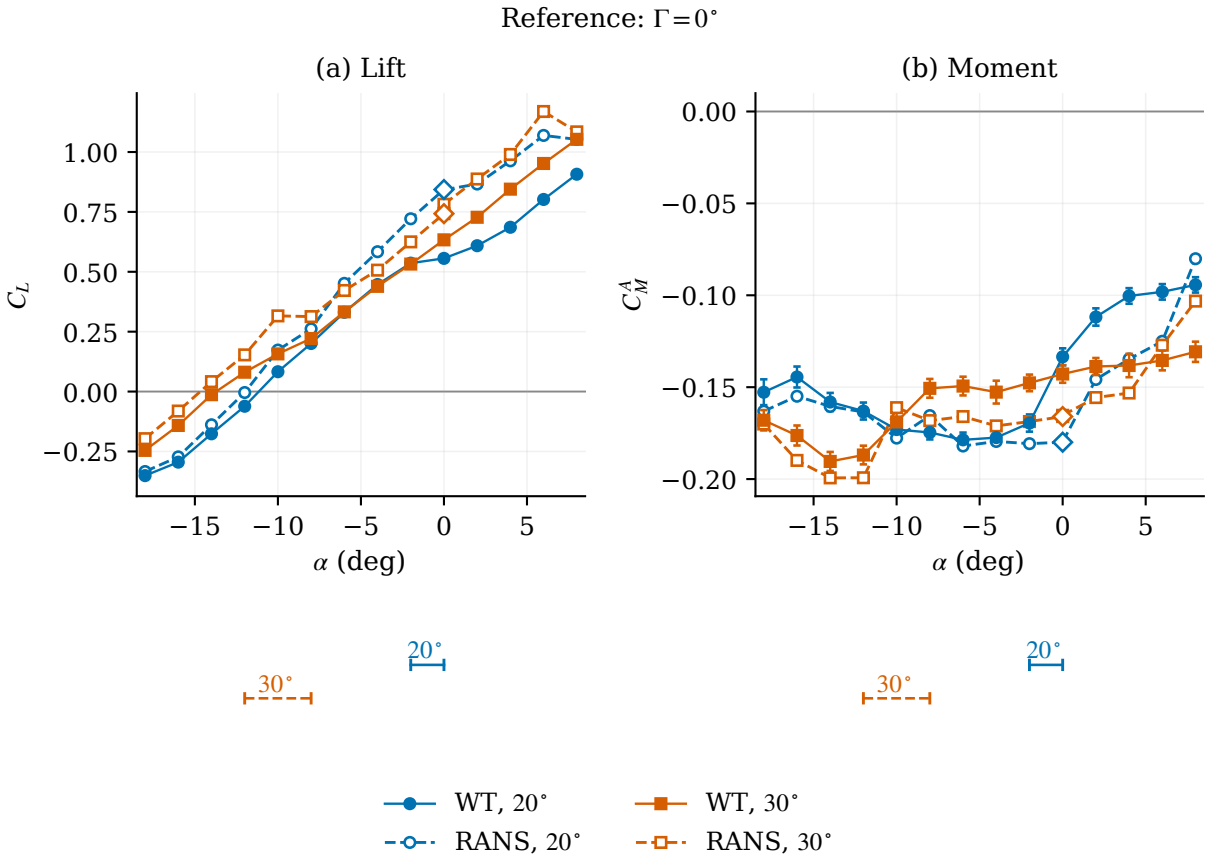


Fig. 4 Reference zero-dihedral configuration: wind-tunnel and RANS lift and pitching moment at two control deflections.

Table 3 Measured lift and moment departures with their uncertainty sensitivities.

Geom., Γ, δ	Coefficient	Interval (deg)	D	u_{ind}	u_{L1}
I, 0°, 20°	C_L	$-2 \rightarrow 0$	-0.08274	0.02357	0.03631
I, 0°, 20°	C_M^A	$-2 \rightarrow 0$	+0.03148	0.00862	0.01360
I, 45°, 20°	C_L	$-2 \rightarrow 0$	-0.04785	0.01694	0.02634
I, 45°, 20°	C_M^A	$-2 \rightarrow 0$	+0.00847	0.02630	0.04082
I, 45°, 30°	C_L	$-2 \rightarrow 0$	-0.04038	0.01646	0.02996
I, 45°, 30°	C_M^A	$-4 \rightarrow -2$	-0.02860	0.02478	0.03822
II, 0°, 20°	C_L	$0 \rightarrow 2$	-0.06902	0.02090	0.03739
II, 0°, 20°	C_M^A	$0 \rightarrow 2$	+0.01668	0.00793	0.01424

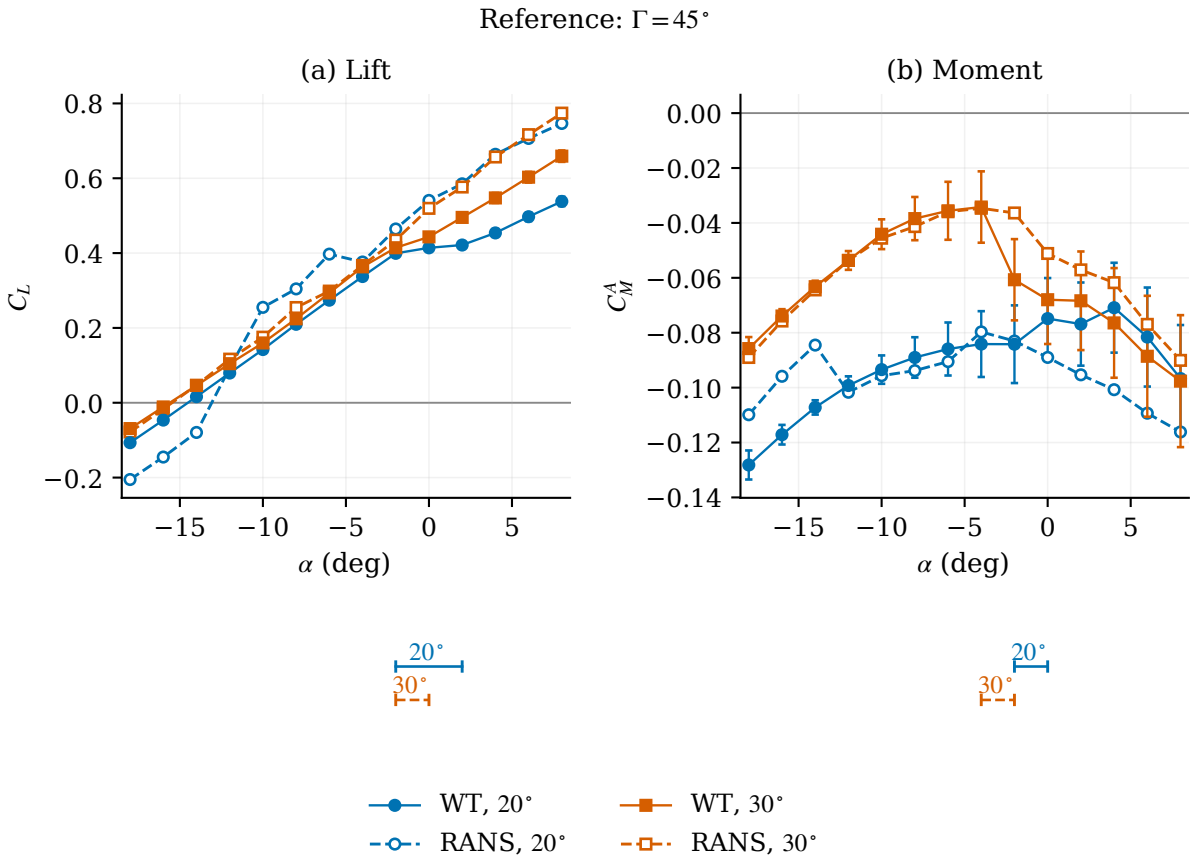


Fig. 5 Reference configuration with positive dihedral: wind-tunnel and RANS lift and pitching moment at two control deflections.

The lower-deflection result shows why the two coefficients should be examined together. At 10°, the measured moment increment changes from −0.0038 over −2 → 0° to +0.011 over 0 → 2°, so a moment-response change is present below the deflection producing the pronounced 20° lift slowdown.

Increasing deflection to 30° instead gives a broader early feature: lift increments are 0.077, 0.064, and 0.111 over −12 → −10, −10 → −8, and −8 → −6°, while moment already increases by about 0.018 on each of the first two intervals. Curvature in the preceding lift sequence makes its departure window-sensitive, spanning both signs. Local increments therefore describe this feature more directly than one departure amplitude.

Introducing 45° positive dihedral changes the joint response. For the reference configuration with 45° dihedral at $\delta = 20°$, measured lift growth remains suppressed across −2 → 2°, but the positive moment increment over −2 → 0° has a departure smaller than both uncertainty sensitivities. At $\delta = 30°$, the selected lift slowdown occurs over −2 → 0°, whereas moment decreases earlier, by 0.027 over −4 → −2°. That moment departure is comparable with u_{ind} and smaller than u_{L1}. Thus coincident, clearly resolved lift and moment events cannot be assumed for the inclined model.

The second planform provides a further contrast in the joint response. For the forward-swept configuration at

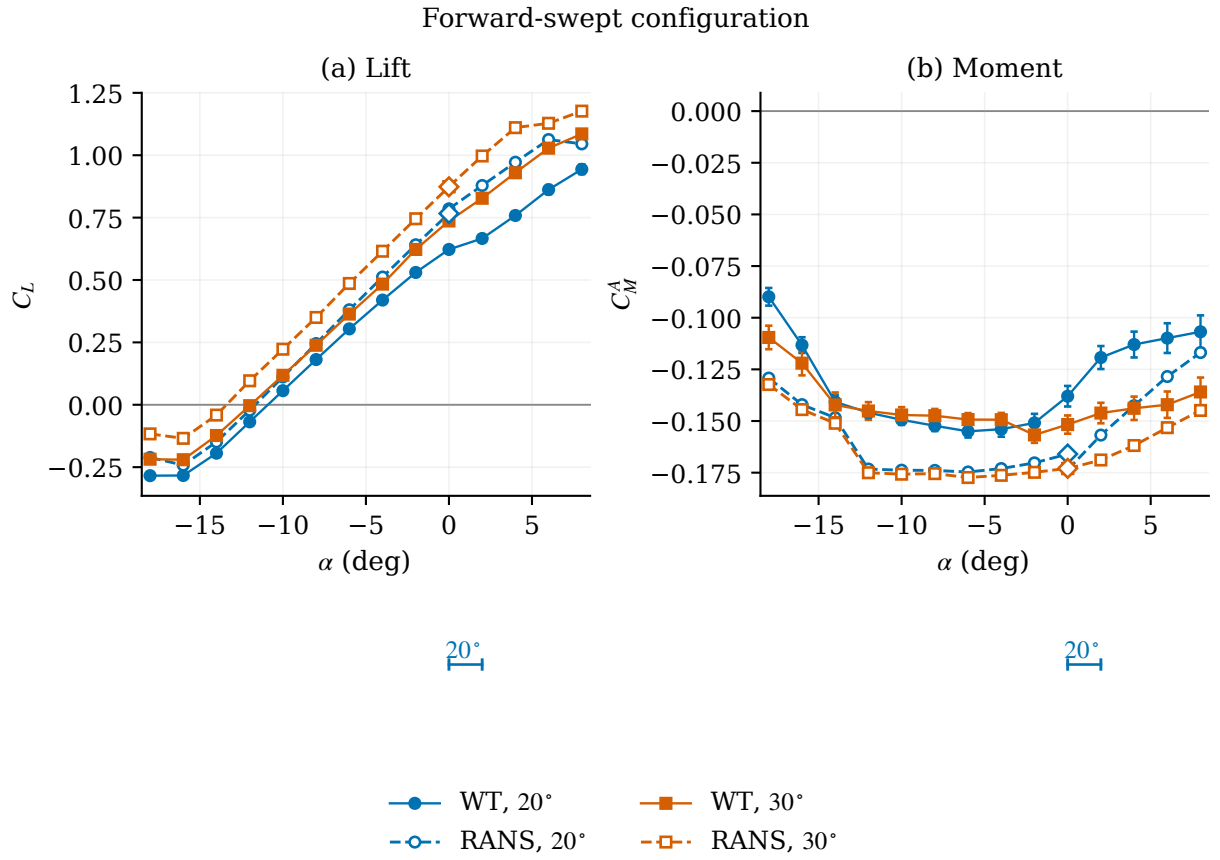


Fig. 6 Forward-swept configuration: wind-tunnel and RANS lift and pitching moment at two control deflections.

$\delta = 20°$, measured lift grows by 0.044 over $0 \rightarrow 2°$, compared with approximately 0.092 on the preceding and following intervals, and moment increases by 0.019. The primary moment departure exceeds both sensitivities, although results obtained using neighboring preceding-incidence fitting ranges do not uniformly exceed the primary L1 sensitivity. Its lift and moment changes are locally evident, with the 30° response also shown in Fig. 6 for context.

To assess what the surface fields can explain, the numerical response is compared separately for lift and moment. For the reference configuration at $\delta = 20°$, the RANS lift slowdown occurs over $0 \rightarrow 2°$, one sampled interval later than in WT. The numerical zero-incidence lift coefficients are approximately 0.842–0.843, compared with WT 0.556 and uncertainty magnitude 0.013. A qualitatively similar slowdown and less-negative moment response therefore coexist with substantial lift-level bias. At 30°, the early numerical lift change is sharper than the broad measured feature, and its fitted-center moment departure is window-sensitive.

At $\Gamma = 45°$, numerical $\delta = 20°$ lift decreases over $-6 \rightarrow -4°$ while WT increases; the computed sequence does not reproduce the sustained measured slowdown or weak moment response convincingly. At $\delta = 30°$, a negative moment change is present, with the strongest numerical bracket later than the measured $-4 \rightarrow -2°$ interval. For the

forward-swept configuration at $\delta = 20^\circ$, the direct numerical $0 \rightarrow 2^\circ$ lift increment is 0.094, more than twice WT 0.044. The corresponding fitted-center moment increment, 0.017, is locally close to WT 0.019, retaining the distinct reference constructions.

The reference calculation reproduces the direction of the joint response, whereas correspondence is weaker for the inclined and forward-swept cases. The numerical load decomposition is therefore used to investigate how changes in surface pressure and the development of locally reversed flow contribute to the computed lift and pitching-moment response.

IV. Joint Numerical Loading Response

A. Ordinary growth and reduced growth of the fixed surface

For the reference RANS sequence, the continuing total lift increase is supplied by a gain on the fixed surface that outweighs a loss on the control (Fig. 7 and Table 4). This occurs for the zero-dihedral reference configuration at $\delta = 20^\circ$ over $0 \rightarrow 2^\circ$. The positive fixed-surface increment, however, is smaller than continuation of its preceding slope would give. Relative to the shared baseline $\{-6, -4, -2\}^\circ$, both components have negative departures and the total is $D_L = -0.111$. The reduction in total lift growth consequently involves the fixed surface as well as the control, despite their opposite ordinary increments.

Table 4 Computed fixed-surface, control-surface, tip, and total increments and departures about the fitted moment references.

Geom., δ, interval	Measure	Fixed	Control	Tip	Total
I, 20°, $0 \rightarrow 2^\circ$	ΔC_L	+0.059287	-0.035104	-0.000004	+0.024179
I, 20°, $0 \rightarrow 2^\circ$	D_L	-0.069179	-0.041428	+0.000004	-0.110605
I, 20°, $0 \rightarrow 2^\circ$	ΔC_M^A	+0.015562	+0.018314	+0.000002	+0.033878
I, 20°, $0 \rightarrow 2^\circ$	D_M^A	+0.010689	+0.022553	-0.000004	+0.033238
I, 30°, $-10 \rightarrow -8^\circ$	ΔC_L	+0.040586	-0.043560	-0.000002	-0.002974
I, 30°, $-10 \rightarrow -8^\circ$	D_L	-0.067327	-0.052888	-0.000002	-0.120216
I, 30°, $-10 \rightarrow -8^\circ$	ΔC_M^A	-0.014959	+0.007907	+0.000002	-0.007050
I, 30°, $-10 \rightarrow -8^\circ$	D_M^A	-0.017339	+0.015008	+0.000002	-0.002328
II, 20°, $0 \rightarrow 2^\circ$	ΔC_L	+0.101449	-0.007930	-0.000010	+0.093509
II, 20°, $0 \rightarrow 2^\circ$	D_L	-0.026344	-0.010408	-0.000003	-0.036755
II, 20°, $0 \rightarrow 2^\circ$	ΔC_M^A	+0.011164	+0.005658	-0.000002	+0.016820
II, 20°, $0 \rightarrow 2^\circ$	D_M^A	+0.007512	+0.007105	-0.000001	+0.014614

Moment about the fitted aerodynamic center gives a different component ranking. Both moment departures are positive, and the control contributes more strongly (Table 4). Thus the fixed surface supplies the larger lift-growth deficit while the control supplies the larger moment departure, even though both are formed from the same loading states. In Fig. 7, dashed segments continue the preceding trends from the direct zero-incidence values, and the shading

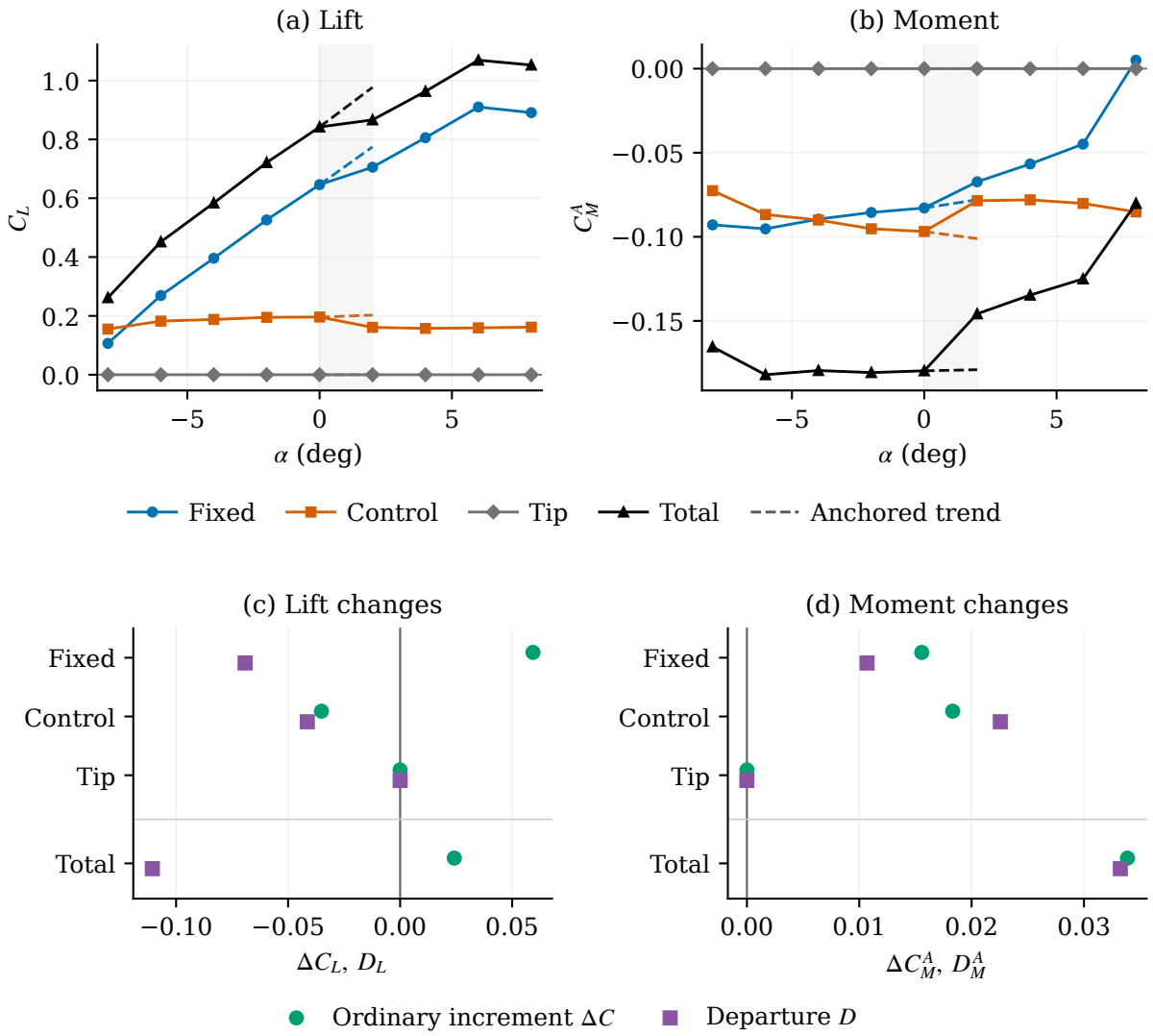


Fig. 7 Fixed-surface, control-surface, and total numerical loading, with ordinary increments and departures from preceding incidence trends.

identifies the $0 \rightarrow 2^\circ$ interval. Their endpoint gaps are the departures in Eq. 3. The difference in component ranking motivates examination of where the pressure changes occur and how they are weighted in moment.

About 63% of the computed lift departure relative to the selected preceding trends is allocated to the fixed surface. Its share remains 62.4–63.0% in the baseline-range and zero-incidence sensitivity checks, while the total departure varies from about −0.137 to −0.105 (Appendix A). Thus the component ranking is more stable than the absolute departure; the percentage describes the computed budget under those definitions.

B. Pressure redistribution and spatial moment weighting

The pressure sections test whether the fixed-surface deficit is confined to the hinge neighborhood or involves a broader chordwise redistribution. At the same $\alpha = 0, 2^\circ$ states used in the budget, all three stations show stronger upper leading-edge suction and a weaker hinge suction minimum (Fig. 8). Circles/squares distinguish incidence, solid/dashed curves the upper/lower sides, and the dotted line locates the hinge. Here, pressure recovery denotes the chordwise rise downstream of the hinge suction minimum. The two-state sections show ordinary pressure changes; departures additionally require the preceding-incidence integrations.

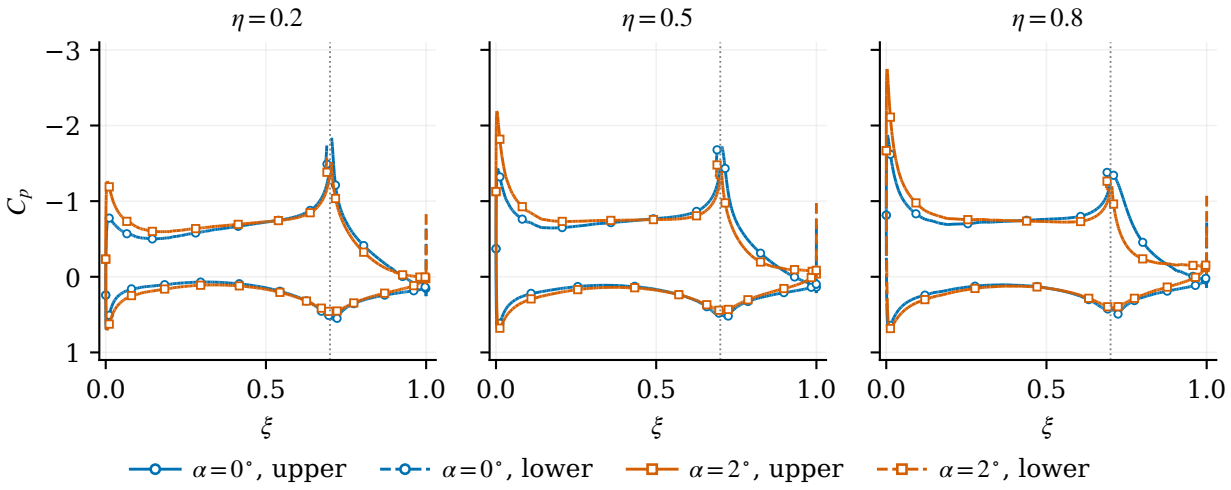


Fig. 8 Chordwise pressure distributions at three span locations across the reference pre-stall loading change.

The regional integrations provide that distinction (Fig. 9). The forward region still gains lift, but less than its preceding trend, while the intermediate region loses lift. Both regions forward of the hinge therefore contribute to the fixed-surface departure. The aft region also loses lift, giving negative departures in all three chordwise regions. A stronger leading-edge suction peak is thus compatible with reduced overall fixed-surface lift growth when the rest of its loading and its preceding trend are considered.

Pressure accounts for nearly all the principal lift departure. Geometric regional sums and solver fixed/control sums cover the complete surface, with a slightly different assignment of the hinge junction. Their small differences therefore reflect the partition rather than a missing contribution.

Along the aft-swept span, similar aft lift losses give moment increments that grow from about 0.0025 inboard to 0.0091 outboard. Outboard aft loading lies farther downstream of the fitted axis, increasing the streamwise arm $x - x_A$ in Eq. 5. Loss of upward force at a downstream location produces a positive moment change, weighted more strongly as that arm increases. This relates the larger outboard moment contribution to the load location, consistent with finite-wing pitching response [3]. Reference and partition sensitivities are discussed in Appendix A.

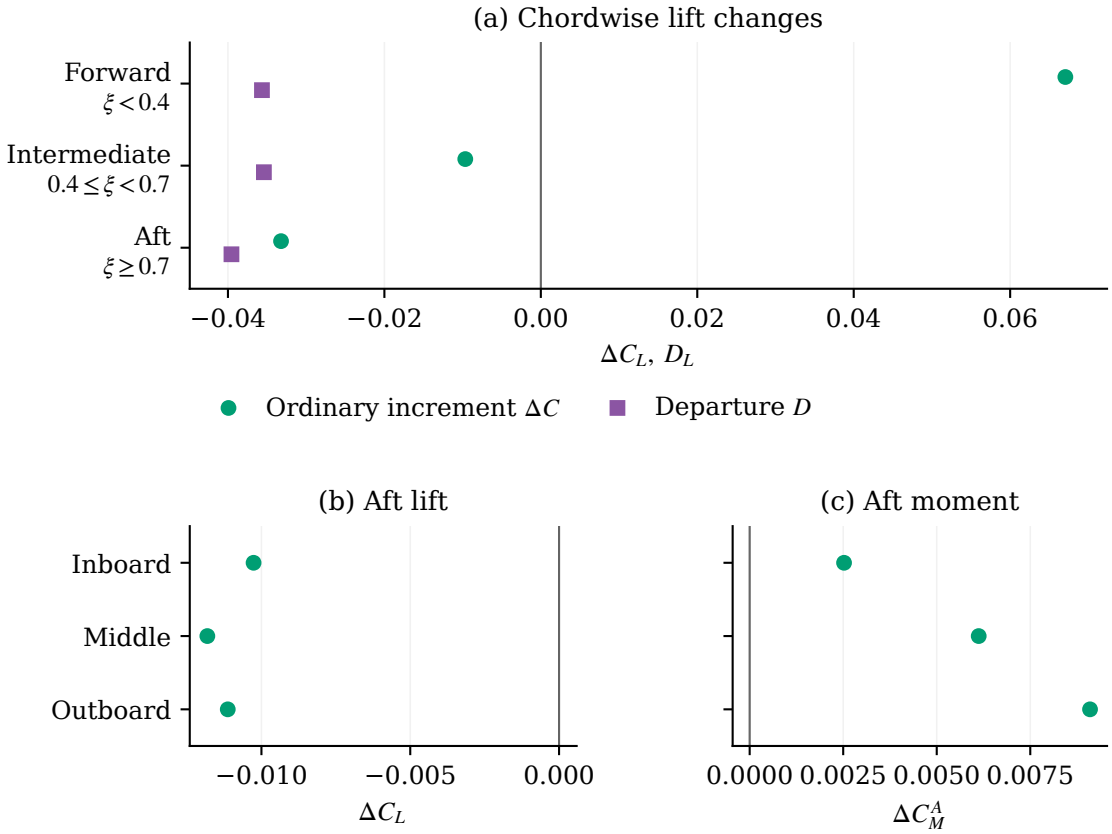


Fig. 9 Chordwise and spanwise contributions to reference lift and pitching-moment changes over the selected pre-stall interval.

C. Control deflection and separated-flow development

Increasing control deflection tests whether the reference distinction between ordinary growth and reduced growth persists in the earlier nonlinear interval. The reference $\delta = 30°$ results in Table 4 again combine positive ordinary fixed lift growth with negative fixed and control departures. The preceding range for this early interval is $\{-16, -14, -12\}°$. Its fitted-center moment contributions compete, however, and the total departure changes sign across tested windows. The lift-budget pattern extends to this deflection, while its moment response is more sensitive to the empirical preceding trend.

The associated wall-shear measure relates this budget to the changing aft pressure and loading. For the reference configuration at $\delta = 20°$, the upper-aft reverse-flow fraction rises from 0.146 to 0.485 over $0 \rightarrow 2°$, with local reverse flow already present at $-2°$ (Fig. 10). At $\delta = 30°$, it rises from 0.622 to 0.935 over $-10 \rightarrow -8°$. Expansion of the existing reverse-flow region accompanies reduced aft loading; its area does not represent every form of three-dimensional separation. This growth persists under the near-zero wall-shear thresholds examined in Appendix A.

The inclined solution already has extensive control-surface reverse flow over the selected moment interval. Its later

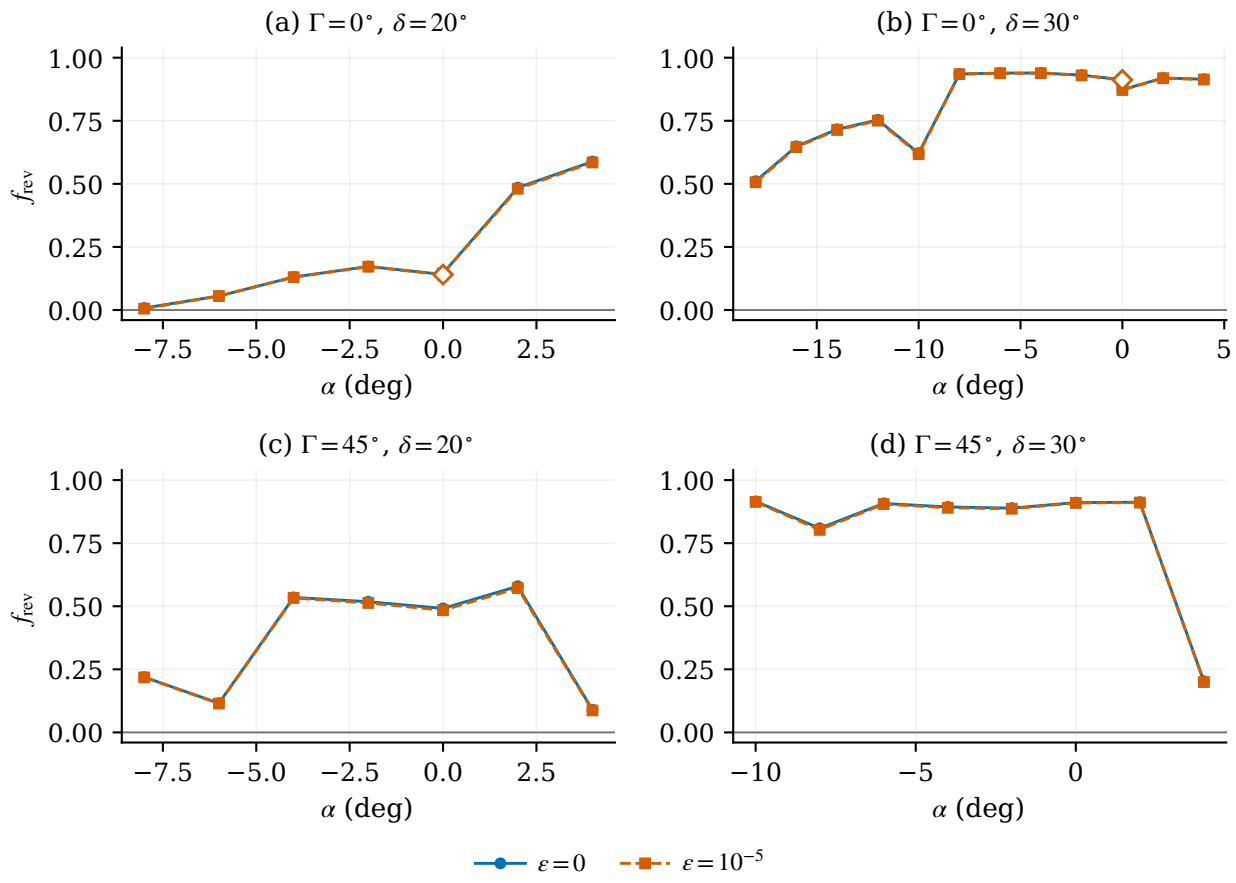


Fig. 10 Extent of reversed chordwise wall shear over the deflected control at different incidences and dihedral settings.

reverse-flow fractions change nonmonotonically with incidence, showing that the extent of reversed wall shear alone does not determine the integrated moment response. The values and state sensitivity are retained in Appendix A; the available steady solutions do not establish a transition or reattachment mechanism.

At $\Gamma = 45°$, $\delta = 30°$, and $\alpha = 2°$, locally reversed chordwise shear also occurs on the upper outboard leading edge ($2/3 \le \eta \le 0.95$, $0 \le \xi \le 0.15$). Figure 11 shows the stored BCM intermittency and signed chordwise skin friction on the upper surface. BCM intermittency in the reversing leading-edge subset has median 2.4×10^{-5} and maximum 8.6×10^{-5}. This algebraic factor ranges from zero to one and controls SA turbulence production; the map identifies the model response associated with local reverse flow. Wall values alone do not establish the off-wall boundary-layer state or a laminar separation bubble.

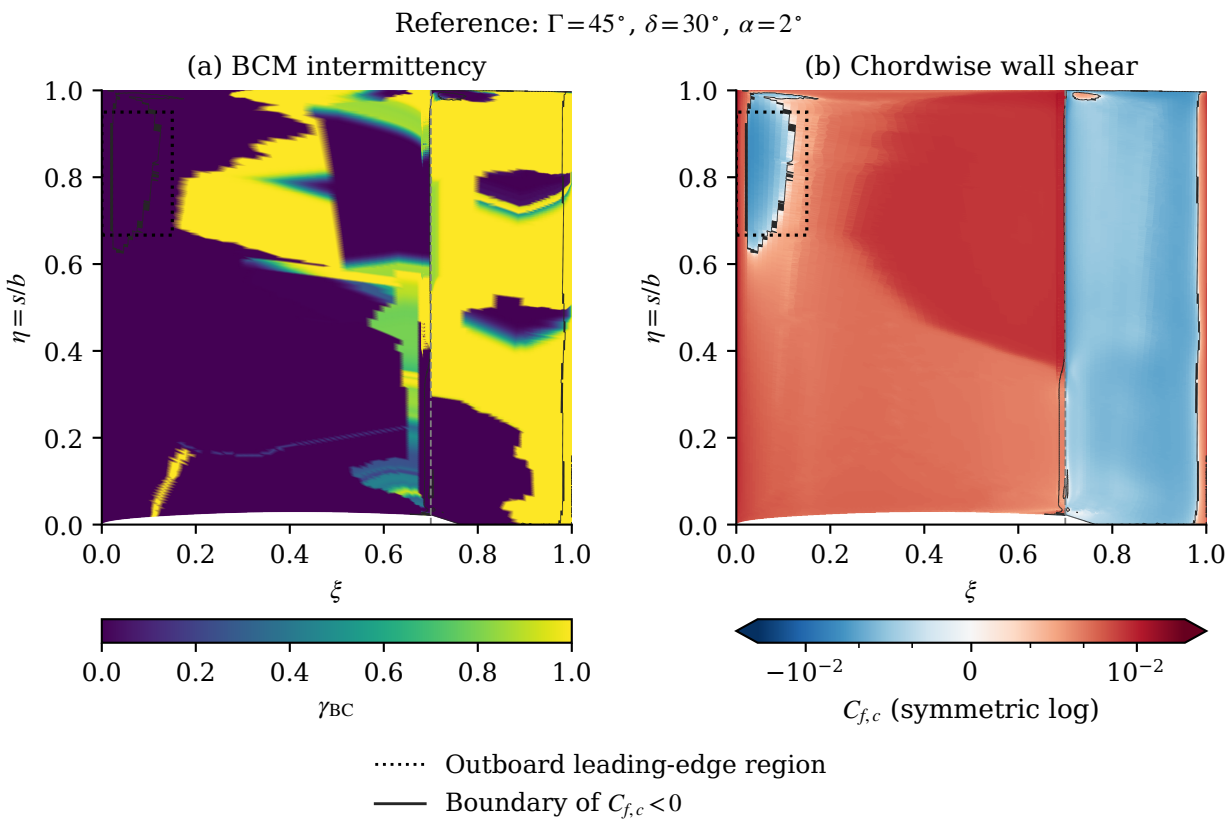


Fig. 11 Inclined reference configuration: upper-surface BCM intermittency and chordwise skin friction at $\delta = 30°$, $\alpha = 2°$.

V. Dihedral and Planform Dependence

A. Matched dihedral: levels and incidence changes

Positive dihedral changes both the projection of aerodynamic forces and the spatial arms through which they contribute to pitching moment. The comparison holds planform, incidence, deflection, area, and MAC fixed, with the associated root cut changing with dihedral. Figure 12 overlays the measured configurations at identical deflections, making the changes in coefficient level and incidence response directly visible.

The measurements first distinguish the size of the lift departure from its size relative to each model's preceding growth. At $\delta = 20°$, absolute departures are smaller with dihedral, but over $-2 \rightarrow 0°$ the descriptive relative deficits are 80.6% and 76.4% for $\Gamma = 0$ and 45°, whereas over $-2 \rightarrow 2°$ they are 64.5% and 82.2%. These sampled ratios have no assigned significance level. A smaller absolute departure consequently does not imply uniformly weaker relative suppression.

The numerical force decomposition addresses the absolute level change before considering its evolution with incidence. At $\alpha = 0°$, $\delta = 20°$, projection of the zero-dihedral normal load contributes about -0.247 of the matched lift

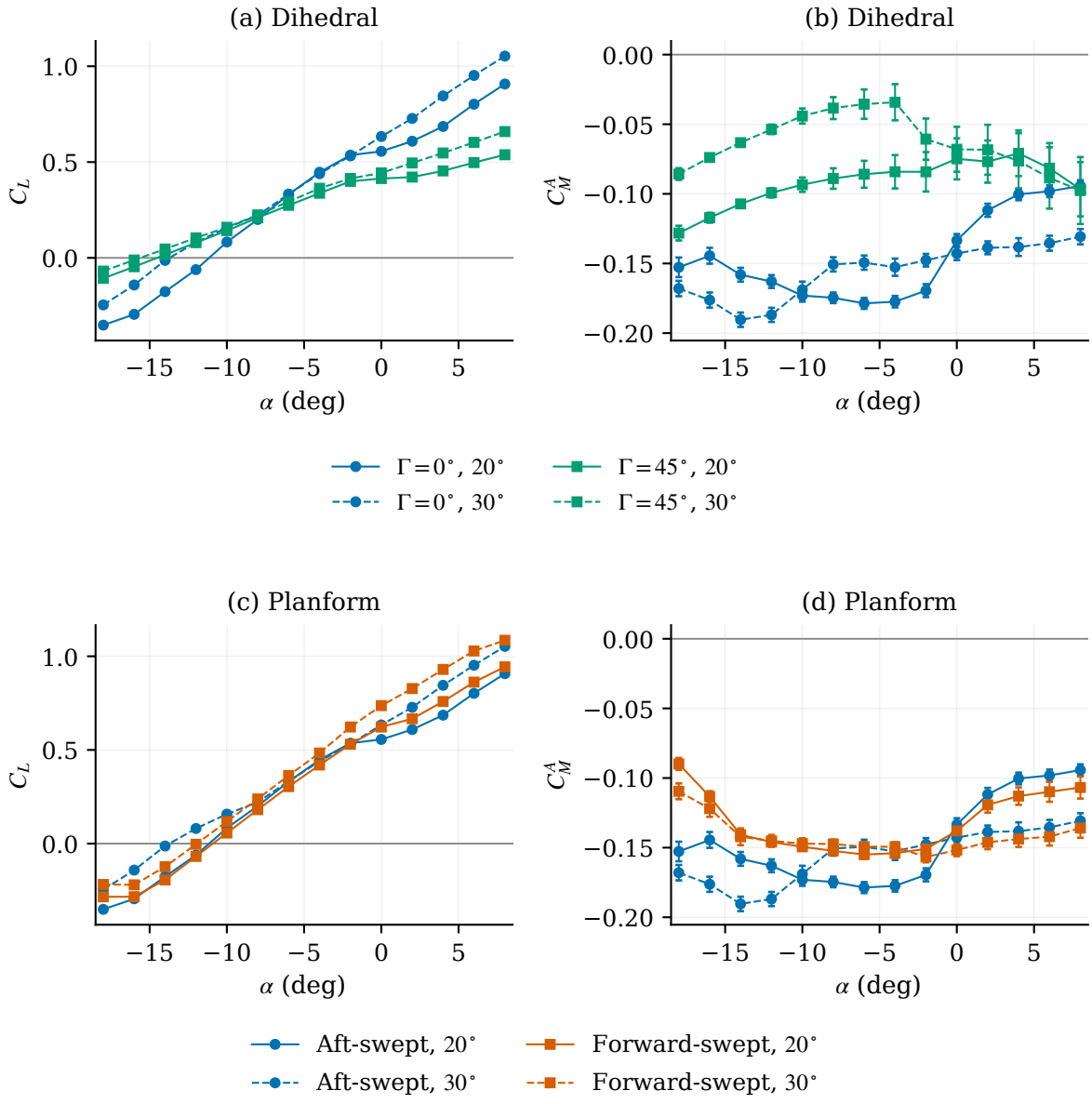


Fig. 12 Matched-deflection wind-tunnel comparisons of lift and pitching moment: dihedral on the reference planform, and aft-swept versus forward-swept planforms.

difference −0.301. Changed local normal loading and span-force projection complete the balance. This coordinate accounting distinguishes projection from changed forces in the level difference, without isolating a causal contribution to nonlinear lift growth.

Moment requires a further comparison because dihedral changes both loading and its spatial weights. To examine how the moment evolves over the same incidence interval, rather than only comparing its absolute level, both numerical configurations are referred to the common axis A_I. Figure 13 evaluates

$$\mathcal{I}_M = [C_{M,45}^{A_I}(\alpha_b) - C_{M,45}^{A_I}(\alpha_a)] - [C_{M,0}^{A_I}(\alpha_b) - C_{M,0}^{A_I}(\alpha_a)] \tag{6}$$

with identical incidence intervals. Contrasts of the integrated $z\,dF_x$ and $-(x - x_{A_I})dF_z$ terms complete this ordinary-increment difference. For $\delta = 30°$ over $-2 \to 0°$, $\mathcal{I}_M \approx -0.018$: the inclined moment increment is more negative than the zero-dihedral increment. Both the axial- and normal-force moment contrasts contribute to that reduction. Their changing contributions across the other intervals show how force changes and spatial moment weighting enter the incidence-dependent response. This comparison uses no baseline subtraction and does not separate geometry from its effect on the computed flow.

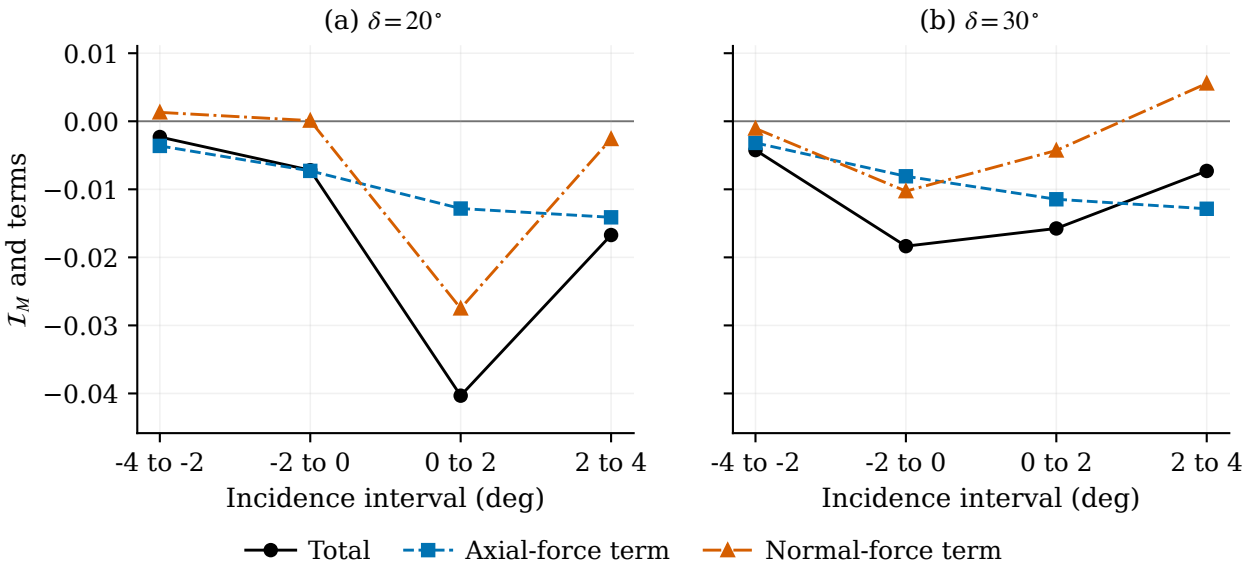


Fig. 13 Dihedral differences in pitching-moment increments, decomposed into axial- and normal-force contributions at matched incidence intervals.

B. Second planform and scope of the result

The forward-swept configuration tests whether the reference component relationship persists with the contrasting planform. Its measured response is overlaid with the aft-swept model in Fig. 12; its WT–RANS correspondence is

shown in Fig. 6. At $\delta = 20°$, Table 4 again gives positive fixed-surface lift growth, negative control growth, and negative departures of both contributions. The fitted-center moment departures are positive and reinforce.

This recurrence concerns the computed response, for which the measured lift slowdown is poorly reproduced. Since taper and aspect ratio also change, the result characterizes the forward-swept planform rather than a sweep-only variation. Reference-point and zero-incidence sensitivities are retained in Appendix A.

The joint budget identifies information not supplied by continuing total lift growth: which surface contributes to the reduced growth and how the location of its load changes affects moment. The present departures vary incidence at fixed deflection, so they do not measure a control derivative with respect to deflection. The result characterizes the tested planforms and full-span control geometry; it does not establish a general failure of reduced-order models.

VI. Conclusions

The reference calculation shows that continued fixed-surface lift growth can coexist with reduced growth of both the fixed and control contributions. At zero dihedral and a 20° deflection, about 63% of the computed lift departure relative to the selected preceding trends is allocated to the fixed surface. The control contributes more strongly to the pitching-moment departure about the fitted center. Stronger leading-edge suction together with reduced intermediate/aft loading, and larger outboard moment arms, connect this different allocation to the spatial loading changes.

The measured response establishes why lift and moment must be considered together: the pronounced 20° lift slowdown is accompanied by a less-negative moment, while a moment-response change is already present at 10°. The earlier 30° lift feature is broader, and the weak inclined moment departures are not clearly resolved above both uncertainty sensitivities.

Matched dihedral comparisons distinguish changes in force projection from changes in incidence-dependent moment contributions. The forward-swept calculation also gives reinforcing fitted-center moment departures, extending the component result to that tested planform.

RANS reproduces the principal reference trends with a displaced lift interval and substantial coefficient-level bias; correspondence is weaker for the inclined and forward-swept responses. Future computations on the finer levels of the established mesh hierarchy are planned to assess the resolution sensitivity of the pre-stall load redistribution, with particular attention to pitching moment and the fixed- and control-surface contributions. The reference result nevertheless identifies a joint loading response that total lift growth alone would conceal.

A. Sensitivity of the Numerical Loading Interpretation

A. Baseline and zero-incidence choices

For the zero-dihedral reference configuration at $\delta = 20^\circ$ over $0 \rightarrow 2^\circ$, Table 5 separates the absolute departure from its component allocation. The three unweighted windows use identical samples for every component, and direct and symmetry-reflected zero-incidence solutions are retained separately. Although the earliest pre-feature window has appreciable curvature, the positive fitted-center moment departure and fixed lift fraction persist as the total lift deficit varies.

Table 5 Reference loading departures over $0 \rightarrow 2^\circ$: sensitivity to the preceding-incidence range and zero-incidence solution.

Zero solution	Baseline (deg)	Total D_L	Fixed fraction (%)	Total D_M^A
Direct	$\{-6, -4, -2\}$	-0.110605	62.55	+0.033238
Direct	$\{-8, -6, -4\}$	-0.136788	62.43	+0.040913
Direct	$\{-4, -2, 0\}$	-0.105014	62.64	+0.033958
Reflected	$\{-6, -4, -2\}$	-0.111790	62.76	+0.033493
Reflected	$\{-8, -6, -4\}$	-0.137973	62.60	+0.041168
Reflected	$\{-4, -2, 0\}$	-0.106792	62.98	+0.034341

The early reference $\delta = 30^\circ$ fitted-center total moment departure varies from approximately -0.0262 to $+0.0076$ across tested windows. For the forward-swept configuration at $\delta = 20^\circ$, the symmetry-reflected zero-incidence solution gives a $0 \rightarrow 2^\circ$ lift increment of 0.1121 instead of direct 0.094 and a fitted-center moment departure of 0.0070 instead of 0.0146. Each coefficient and surface field retains its corresponding state; alternate zero solutions are not averaged and do not constitute a hysteresis measurement.

B. Spatial partition and reversal threshold

Changing the regional boundaries tests whether the fixed-surface departure depends on a particular partition. The complete fore/aft sums persist when span thirds are replaced by quarters, and finer chord subdivisions preserve the two negative fixed-chord departures. The inboard aft departure changes from about -0.0127 to -0.00939 when its boundary moves from $\eta < 1/3$ to $\eta < 1/4$, illustrating the dependence of a regional magnitude on its definition.

For the reference configuration at $\delta = 20^\circ$, the upper-aft reversal fraction at direct zero is 0.14578 versus 0.14089 at reflected zero. Threshold 10^{-5} changes the direct $0 \rightarrow 2^\circ$ sequence to 0.14497–0.48043, preserving its growth. Typical stored vector skin-friction norms are of order 10^{-3}, so the small thresholds assess near-zero sensitivity without a statistical uncertainty interpretation.

C. Moment-reference effects

At the original numerical origin, the baseline total moment departure is +0.10028, with fixed +0.05286 and control +0.04743. At the fitted aerodynamic center, the corresponding values are +0.03324, +0.01069, and +0.02255. The allocation changes while reinforcement persists. All reference transports use body $C_{F,z}$ in Eq. 2; the experimental comparison retains its own fitted center.

About the original numerical origin, the direct II fixed/control moment departures are $-0.00231/+0.00352$, with total +0.00121. This small cancellation is reference-specific and does not describe fitted-center reinforcement. For the baseline aft-span regions, the moment increments at the original origin are approximately $0.0082, 0.0129, 0.0156$ from inboard to outboard. Their magnitudes differ from those at the fitted center, while the outboard weighting persists.

D. Computed-state sensitivity

The inclined reverse-flow fractions are nonmonotonic: between 2 and 4° they fall from approximately 0.58 to 0.09 at $\delta = 20°$, and from 0.91 to 0.20 at $\delta = 30°$. These are changes between stored steady states, whose physical origin remains unresolved. For the inclined $\delta = 20°$ lift irregularity over $-6 \rightarrow -4°$, terminal lift ranges of order 10^{-4} are much smaller than the between-state change, but do not establish uniqueness of the steady solution.

The interpreted surface fields use their corresponding solver force reports. For the reference $\delta = 30°$, $\alpha = -6°$ point and forward-swept $\delta = 20°$ points at $\alpha = -4°$ and direct 0°, retained terminal histories differ from those reports. The surface/force agreement supports the adopted load accounting, while terminal registration remains incomplete. The reference 20° component result uses matched states; the forward-swept budget retains this provenance limitation.

Acknowledgments

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